Retail Sales Intelligence & Customer Behavior Analytics

Problem Statement:

Retail businesses often struggle to understand customer purchasing behavior, identify key revenue drivers, and optimize strategies due to fragmented data. This project aims to address these challenges by building a comprehensive analytics solution.

1. Project Overview

This project focuses on analyzing retail customer purchase behavior using transactional data to generate actionable business insights.
The objective is to build a **Retail Sales Intelligence Dashboard** that helps businesses:

- Understand **customer purchasing patterns**
- Identify **key revenue drivers**
- Improve **customer engagement and retention**
- Optimize **marketing and pricing strategies**

The project combines **Python, MySQL, and Power BI** to create an end-to-end analytics pipeline.

Tools & Technologies Used:

- Python (Pandas, NumPy, Seaborn, Matplotlib)
- MySQL (Data querying and analysis)
- Power BI (Dashboard and visualization)
- Jupyter Notebook (EDA and analysis)

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2. Dataset Summary

- **Source:** Kaggle – Customer Purchase Behaviour Dataset
- **Records:** ~3,900 transactions
- **Columns:** 18

Key Features:

- **Customer Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item Purchased, Category, Purchase Amount, Season
- **Behavioural Data:** Discount Applied, Review Rating, Shipping Type
- **Customer History:** Previous Purchases, Frequency

Data Quality Issues:

- Missing values in **Review Rating (~37 records)**
- Redundant column: **promo_code_used**

3. Data Preparation & Feature Engineering (Python)

We performed data cleaning and transformation using Python.

3.1 Data Loading & Exploration

- Loaded dataset using Pandas
- Used .info() and .describe() to understand structure
- Identified missing values

3.2 Data Cleaning

- Imputed missing **review_rating** using **category-wise median**
- Standardized column names (snake_case)
- Removed duplicate records
- Dropped redundant column (**promo_code_used**)

3.3 Feature Engineering

To enhance analysis, new features were created:

Age Group Segmentation

- Teen
- Young Adult
- Adult
- Mid Age
- Senior

Customer Segmentation

Based on purchase history:

- **New:** ≤ 2 purchases
- **Returning:** 3–5 purchases
- **Loyal:** > 5 purchases

Revenue Column

- Created revenue = purchase_amount

3.4 Data Export

- Exported cleaned dataset to:
  - CSV file
  - MySQL database (customers table)

4. Exploratory Data Analysis (Python)

We performed visual analysis to identify trends and patterns.

4.1 Customer Analysis

- Revenue Distribution Across Customer Age Groups
- Revenue Distribution by Customer Segments
- Impact of Subscription Status on Customer Revenue by Segment

4.2 Product & Category Analysis

- Purchase Amount Distribution Across Product Categories
- Top 10 Revenue-Generating Products

4.3 Pricing & Discount Analysis

- Revenue Distribution: Discount Vs Non-Discount Transactions
- Impact of Discounts on Revenue Distribution

4.4 Sales & Time-Based Trends

- Seasonal Trends in Revenue Performance

4.5 Customer Experience Analysis

- Distribution of Customer Review Ratings
- Relationship Between Customer Ratings and Revenue

4.6 Correlation & Statistical Analysis

- Correlation & Relationship Analysis Among Key Variables
- Correlation Matrix of Key Numerical Features

5. Data Analysis Using SQL (Business Queries)

Structured analysis was performed using MySQL to simulate real-world business scenarios.

5.1 BASIC QUERIES

- Q1. What is the total revenue generated?

- Q2. What is the revenue distribution by gender?
- Q3. Which age groups contribute the most to revenue?
- Q4. Which product categories generate the highest revenue?

Result Grid | Filter Rows:

total_revenue
233081

gender	total_revenue
Male	157890
Female	75191

Result Grid | Filter Rows:

age_group	total_revenue
Senior	88480
Mid Age	65629
Adult	44342
Young Adult	30491
Teen	4139

Result Grid | Filter Rows:

category	total_revenue
Clothing	104264
Accessories	74200
Footwear	36093
Outerwear	18524

5.2 PRODUCT ANALYSIS

- Q5. What are the top 5 products by total sales?
- Q6. Which products have the highest average ratings?
- Q7. What are the top 3 products within each category based on sales?

Result Grid | Filter Rows:

item_purchased	total_sales
Blouse	10410
Shirt	10332
Dress	10320
Pants	10090
Jewelry	10010

Result Grid | Filter Rows:

item_purchased	avg_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

category	item_purchased	total_sales
Accessories	Jewelry	10010
Accessories	Sunglasses	9649
Accessories	Belt	9635
Clothing	Blouse	10410
Clothing	Shirt	10332
Clothing	Dress	10320
Footwear	Shoes	9240
Footwear	Sandals	9200
Footwear	Boots	9018
Outerwear	Coat	9275
Outerwear	Jacket	9249

5.3 CUSTOMER ANALYSIS

- Q8. How are customers distributed across different segments?
- Q9. How many repeat buyers (more than 5 purchases) exist?
- Q10. What is the total revenue contributed by loyal customers?

Result Grid | Filter Rows: | Expo

customer_segment	customer_count	total_revenue
Loyal	3476	207389
New	155	9197
Returning	269	16495

Result Grid | Filter Rows:

repeat_buyers
3476

Result Grid | Filter Rows:

loyal_revenue
207389

5.4 BEHAVIOR ANALYSIS

- Q11. How does discount application impact revenue and order volume?
- Q12. Which customers used discounts but still spent above average?
- Q13. Do subscribed customers spend more than non-subscribers?

Result Grid | Filter Rows: | Export:

discount_applied	total_orders	avg_revenue	total_revenue
Yes	1677	59.28	99411
No	2223	60.13	133670

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscription_status	shipping_type	discou
2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes
3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes
4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes
7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping	Yes
9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes	Express	Yes
12	30	Male	Shorts	Clothing	68	Hawaii	S	Olive	Winter	4.9	Yes	Store Pickup	Yes
13	61	Male	Coat	Outerwear	72	Delaware	M	Gold	Winter	4.5	Yes	Express	Yes
16	64	Male	Skirt	Clothing	81	Rhode Island	M	Teal	Winter	2.8	Yes	Store Pickup	Yes
20	66	Male	Pants	Clothing	90	Rhode Island	M	Green	Summer	3.3	Yes	Standard	Yes
22	31	Male	Pants	Clothing	62	North Carolina	M	Charcoal	Winter	4.1	Yes	Store Pickup	Yes
24	31	Male	Pants	Clothing	88	Oklahoma	XL	White	Winter	4.4	Yes	Express	Yes

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

subscription_status	customers	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

5.5 SALES & TREND ANALYSIS

- Q14. Which season generates the highest revenue?
- Q15. How does shipping type affect order volume and average order value?
- Q16. Which locations contribute the most to sales?

Result Grid | Filter Rows:

season	total_revenue
Fall	60018
Spring	58679
Winter	58607
Summer	55777

Result Grid | Filter Rows:

shipping_type	total_orders	avg_order_value
Express	646	60.48
Free Shipping	675	60.41
Next Day Air	648	58.63
Standard	654	58.46
2-Day Shipping	627	60.73
Store Pickup	650	59.89

Result Grid | Filter Rows:

location	total_revenue
Montana	5784
Illinois	5617
California	5605
Idaho	5587
Nevada	5514
Alabama	5261
New York	5257
North Dakota	5220
West Virginia	5174
Nebraska	5172
New Mexico	5014
Minnesota	4977

5.6 ADVANCED INSIGHTS

- Q17. Which categories have the highest average product ratings?
- Q18. Which products rely most on discounts (highest discount percentage)?
- Q19. Which age group has the highest average spending?

Result Grid | Filter Rows:

category	avg_rating
Footwear	3.79
Accessories	3.77
Outerwear	3.75
Clothing	3.72

Result Grid | Filter Rows:

item_purchased	discount_percentage
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Result Grid | Filter Rows:

age_group	avg_spending
Young Adult	60.74
Teen	59.99
Senior	59.95
Adult	59.76
Mid Age	59.07

6. Dashboard Development (Power BI)

An interactive dashboard was created to visualize insights.

6.1 Dashboard Pages

Retail Sales & Customer Intelligence Overview

- Revenue by Category
- Revenue by Gender
- Customer Segments Contribution
- Revenue Trend Over Season
- Revenue by Age Group
- Average Orders Per Customer
- Key Insights

6.2 Product & Customer Behavior Analysis

Product Analysis

- Top Revenue Generating Products
- Category Distribution
- Product Ratings Overview

Customer Behavior

- Subscription Vs Non Subscription
- Discount Impact
- Customer Spending Behavior

Operational Analysis

- Shipping Type Analysis
- Purchase Frequency

6.3 Revenue Driver & Regional Analysis

- Revenue Driver Analysis (Decomposition Tree)
- Revenue Distribution by Location

6.4 Filters

- Category
- Gender
- Age Group
- Season
- Subscription Status
- Shipping Type
- Location

7. Key Business Insights

- Loyal customers contribute the majority of total revenue, highlighting strong retention impact
- Clothing category consistently drives the highest sales across all segments
- A small group of products generates a significant share of overall revenue
- Seasonal trends (Winter and Fall) show peak revenue performance
- Discounts increase purchase activity but should be optimized for profitability

- Customer segments differ in both contribution and spending behavior
- Certain locations contribute disproportionately to total revenue, indicating regional opportunities

8. Business Recommendations

A. Customer Strategy

- Focus on retaining loyal customers through loyalty programs and personalized offers
- Identify high-value customers and target them with premium product recommendations
- Improve engagement for new and regular customers to convert them into loyal customers
- Use customer segmentation to tailor communication and improve customer lifetime value

B. Marketing Strategy

- Invest more in marketing high-performing categories like Clothing
- Run targeted campaigns during peak seasons (Winter/Fall) to maximize revenue
- Use personalized marketing based on customer behavior and purchase history
- Focus on high-performing regions with localized promotions and campaigns

C. Pricing Strategy

- Use discounts strategically to increase conversions without reducing margins
- Identify customers who purchase without discounts and avoid unnecessary offers
- Implement dynamic pricing during high-demand seasons
- Analyze discount effectiveness regularly to optimize pricing decisions

D. Product Strategy

- Focus inventory and promotions on top-performing products (e.g., Shirts, Blouses)
- Improve visibility and availability of high-demand categories
- Review low-performing products and optimize or phase them out
- Leverage customer reviews and ratings to improve product offerings

9. Deliverables

1. **Data Preparation & Modeling (Python)**
    - Cleaned and transformed dataset
    - Created new analytical features
2. **Data Analysis (SQL)**
    - Developed 18+ business queries
    - Extracted actionable insights
3. **Visualization (Power BI)**
    - Built interactive dashboard
    - Enabled real-time decision-making

4. **Project Report & Presentation**
    - Documented insights and recommendations
5. **GitHub Repository**
    - Included Python, SQL, and dashboard files

10. Conclusion

This project successfully developed an end-to-end Retail Sales Intelligence and Customer Behavior Analytics solution by integrating data processing, analysis, and visualization. Using Python for data preparation, MySQL for business-driven querying, and Power BI for interactive dashboards, the project transformed raw transactional data into meaningful insights.

The analysis identified key revenue drivers such as loyal customers, high-performing product categories, and seasonal demand patterns. It also highlighted the impact of customer behavior, pricing strategies, and geographic distribution on overall business performance.

The interactive dashboard enables stakeholders to explore data at both high-level and granular levels, supporting data-driven decision-making. The business recommendations derived from the analysis provide practical strategies to improve customer retention, optimize pricing, and enhance product performance.

Overall, this project demonstrates the ability to combine technical skills with business understanding to deliver actionable insights and scalable analytics solutions.